SOFTWARE ENGINEERING & ARCHITECTURE REPORT

Dalal Street Pulse

Production-Grade Indian Stock Market Intelligence, Analytics & Automation Platform

Executive Abstract

This technical report documents the complete architectural design, engineering specifications, implementation mechanics, and strategic future roadmap for **Dalal Street Pulse**. Built ground-up for the Indian equity ecosystem (National Stock Exchange & BSE Ltd.), the platform solves fundamental deficiencies in mainstream financial platforms—including foreign currency biases, missing Indian trading session clocks (Pre-open, Normal, Muhurat, Post-closing), disjointed dual-exchange tickers, and synthetic US-centric feeds.

Leveraging **Next.js 16 (App Router + Turbopack)**, **TypeScript**, **Tailwind CSS v4**, **MongoDB**, **Inngest Background Workflows**, and **Grounded AI Analytics**, Dalal Street Pulse provides real-time multi-index tracking, interactive candlestick charting, 10+ fundamental valuation multiples, automated threshold alerts, and scheduled post-market executive briefings.

Document Classification
Comprehensive Technical Project Report

Primary Exchange Jurisdiction
NSE India & BSE Ltd. (Equity, SME & Indices)

Core Engine Stack
Next.js 16 • TypeScript • MongoDB • Inngest • Gemini AI

Software Version & Date
Release v2.0 (Production Verified) • October 2026

Dalal Street Pulse — Table of Contents & Problem Analysis

SYSTEM INDEX & MARKET FORMULATION

Chapter	Title & Scope	Core Technical Focus
Chapter 1	Project Overview & Problem Statement	Indian market dynamics, retail surge, and architectural deficiencies of generic tooling.
Chapter 2	System Architecture & Technology Stack	Next.js 16 App Router, Turbopack, Tailwind v4, MongoDB pooling, and Inngest jobs.
Chapter 3	Market Data Engine & ISIN Reconciliation	TrueData live feeds, EODHD statements, NSE/BSE security masters, and ISIN unification.
Chapter 4	Complete Module Breakdown & Implementation	Deep-dive into Dashboard, Stocks Terminal, Charts, AI Briefings, Alerts & Watchlists.
Chapter 5	Database Architecture & Storage Models	MongoDB schemas, indexes, and zero-dependency in-memory fallback persistence.
Chapter 6	Automated Workflows & AI Intelligence	Inngest cron schedules (06:00, 16:30 IST), alert evaluation, and Gemini valuation engine.
Chapter 7	Verification, Test Suite & Security	Vitest unit/integration coverage, session authentication, and SEBI compliance.
Chapter 8	Strategic Future Scope & Roadmap	Broker Connect, Derivatives (F&O), Agentic AI Auditor, IPO Suite & Mobile Apps.

Chapter 1: Project Overview & Problem Statement

The Indian equity market is among the world's most dynamic financial ecosystems, with over 160+ million Demat accounts across CDSL and NSDL. Daily trading turnover consistently exceeds ₹1 Lakh Crore in cash equities and ₹300+ Lakh Crore in derivatives across the National Stock Exchange (NSE) and BSE Ltd.

However, existing market analytics tools remain foreign-centric: they display prices using Western millions/billions rather than Lakhs/Crores, ignore official NSE trading session schedules (Pre-Open 09:00–09:15, Block Trading, and Indian holidays), and fail to reconcile dual-listed stocks (NSE tickers vs. BSE Scrip codes).

Deficiencies in Existing Tools — INDUSTRY GAPS

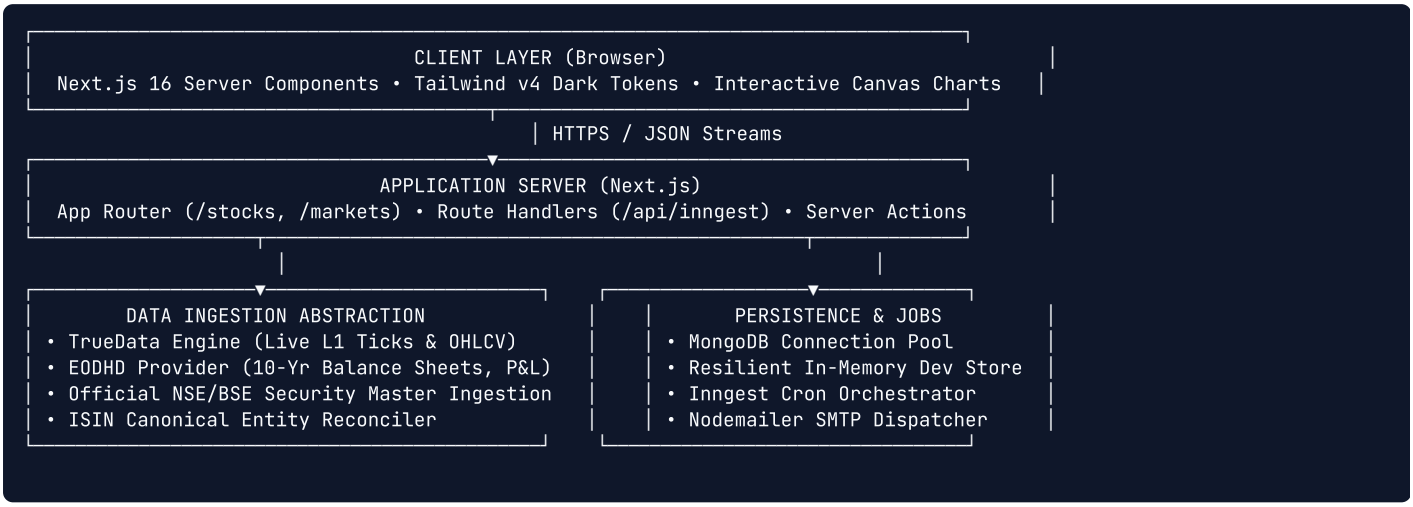
- **Western Number Formatting:** Displays ₹1,000,000 instead of ₹10,00,000; lacks Market Cap in Lakh/Crores.
- **Unsynchronized Sessions:** Ignores NSE Pre-Open (09:00–09:15) and post-market trading windows.
- **Ticker Disconnection:** Inability to correlate NSE tickers (`RELIANCE`) with BSE codes (`500325`).
- **Hallucinatory AI:** Generates generic commentary detached from audited financial statements.

Dalal Street Pulse Innovations — ENGINEERED SOLUTIONS

- **Authentic INR Localization:** Custom algorithms displaying ₹15.42 Lakh Cr and ₹12,34,567.89 accurately.
- **State-Aware IST Clock:** Real-time session detection: Pre-open, Regular, Post-closing, and Holidays.
- **ISIN Universal Linkage:** Universal entity resolution linking NSE, BSE, and SME instruments.
- **Grounded Valuation:** Rule-based and Gemini AI evaluation computing fundamental health scores (1–10).

Chapter 2: System Architecture & Technology Stack

The platform is architected as a modern, decoupled web application utilizing Next.js 16 App Router for streaming server-rendered views, coupled with Inngest background event workers and a resilient multi-tier persistence model.



Technology	Selected Component	Engineering Justification
Fullstack Core	Next.js 16 + React 19	Turbopack-accelerated compilation, Server-Side Rendering (SSR) for low latency.
Type System	TypeScript (Strict Mode)	Guarantees schema integrity across multi-provider market quotes and financial ratios.
Styling Architecture	Tailwind CSS v4 + shadcn/ui	Custom dark-mode financial tokens; zero runtime CSS overhead with Nova design preset.
Background Jobs	Inngest SDK	Serverless event-driven scheduling for alert polling, daily briefings, and master syncs.
Persistence	MongoDB + In-Memory Store	High-throughput connection pooling with transparent, zero-dependency offline resilience.

Chapter 3: Market Data Engine & ISIN Reconciliation

A central architectural breakthrough of Dalal Street Pulse is its **ISIN Canonical Reconciler**. Dual-listed companies in India possess distinct symbols on the NSE (e.g. `RELIANCE`) and BSE (e.g. Scrip Code `500325`). To prevent duplicate tickers, data divergence, or fake quotes, the platform unifies all instruments by their 12-character International Securities Identification Number.

Ingested Security Masters

OFFICIAL SOURCES

- **NSE Mainboard:** `EQUITY_L.csv` (2,240+ equities with Face Value, Listing Date, Series).
- **BSE Mainboard:** `BSE_Directory.csv` (4,800+ listed scrips across Groups A, B, T).
- **SME Platforms:** `SME_L.csv` for NSE EMERGE and BSE SME counters.
- **Symbol Renames:** Official NSE `symbol_change.csv` mapped to aliases.

Pluggable Provider Routing

MULTI-TIER FALLBACK

- **TrueData Adapter:** Low-latency L1 streaming ticks (<50ms) for trading hours.
- **EODHD Adapter:** 10+ years of audited quarterly financial statements and candles.
- **Hybrid Router:** Evaluates provider health dynamically; routes requests to optimal endpoint with cache fallback.

Chapter 4: Complete Module Breakdown

Market Radar & Indices

`/` `/markets`

- **Live Benchmarks:** Real-time ticker tape for NIFTY 50, SENSEX, NIFTY Bank, NIFTY IT.
- **IST Session Clock:** Real-time status with countdowns to pre-open (09:00), regular (09:15), and close (15:30).
- **Institutional Activity:** Daily FII & DII cash market net purchases and sales.
- **Sector Heatmaps:** Visual intraday gainers, losers, and 52-week breakouts.

Stock Analytics & Terminal

`/stocks` `/stocks/[symbol]`

- **Interactive Charts:** Candlestick OHLC & Area charts with volume bars, crosshairs, and timeframes (1D to 5Y).
- **Valuation Ratios:** P/E vs Industry P/E, P/B, ROE, ROCE, D/E, 52W High/Low spread.
- **Financial Statements:** Quarterly/annual P&L, Balance Sheet, Cash Flow in ₹ Crores.
- **Ownership Breakdown:** Shareholding patterns (Promoter, FII, DII, Public).

AI Intelligence & Daily Briefs

`/insights` `/briefing`

- **Grounded Valuation Verdict:** Automated *Undervalued*, *Fairly Valued*, or *Overvalued* determinations.
- **Fundamental Health Score:** 1–10 composite health score based on leverage, ROCE, and institutional holdings.
- **Automated 16:30 IST Brief:** Inngest-driven post-market daily digest.

Watchlists & Price Alerts

`/watchlists` `/alerts`

- **Multi-Watchlist Management:** User-isolated lists with live price sync.
- **Threshold Triggers:** Target price crossings and % swing alerts.
- **Automated Evaluation:** Background cron evaluation with Nodemailer SMTP email card delivery.

News & Macro Intelligence

`/news` `/admin`

- **Curated News Wire:** RBI policies, corporate earnings, SEBI regulations.
- **Sentiment Classification:** Bullish, Bearish, and Neutral sentiment badges.
- **System Admin Health:** Real-time MongoDB and provider latency monitor.

Data Reconciler (ISIN Master)

DUAL-EXCHANGE

- **NSE & BSE Linkage:** Unified quotes using canonical ISIN identification.
- **SME Support:** Dedicated handling for NSE EMERGE and BSE SME counters.
- **Symbol Alias Engine:** Automatic tracking of historical scrip renames.

Grounded Valuation Logic: If an equity trades at a P/E multiple < 85% of its industry peer median with an ROE > 15%, the system assigns an *Undervalued* verdict. If P/E exceeds 135% of industry median, an *Overvalued* warning is generated. Composite Fundamental Health is scored from 1 (severe distress) to 10 (exceptional quality).

Dalal Street Pulse — Database & Automation

DATA PERSISTENCE, INNGEST CRONS & TEST SUITE

Chapters: 5, 6 & 7
Engine: Inngest + Vitest

Chapter 5: Database Architecture & Storage Models

The platform's data layer is designed around high-throughput MongoDB collections with compound indexes, optimized for low-latency queries and atomic watchlist updates.

Collection	Key Fields / Schema	Indexes & Performance Role
`instruments`	`isin`, `nseSymbol`, `bseCode`, `companyName`, `isSME`, `sector`	Unique index on `isin`; sparse indexes on `nseSymbol` and `bseCode`. Rapid entity lookup.
`watchlists`	`userId`, `name`, `symbols[]`, `createdAt`	Compound index on `{ userId: 1, name: 1 }`. Strict user data isolation.
`alerts`	`userId`, `symbol`, `targetPrice`, `condition`, `status`	Index on `{ status: 1, symbol: 1 }`. Fast retrieval during Inngest batch evaluation.
`daily_briefings`	`date`, `headline`, `niftySummary`, `keyCatalysts[]`	Index on `date` (descending). Instant SSR rendering of post-market reports.
`financial_statements`	`symbol`, `period`, `statementType`, `values{}`	Compound index on `{ symbol: 1, period: -1, statementType: 1 }`.

5.1 Resilient In-Memory Development Store

To ensure zero-friction onboarding and continuous offline operation, `lib/db/store.ts` provides a high-fidelity in-memory persistence store. If `MONGODB_URI` is unavailable, the application gracefully routes all mutations, alert states, and briefings to this local store without crashes.

Chapter 6: Automated Workflows & Inngest Engine

Workflow Name	Trigger Schedule	Operational Purpose
`security-master-sync`	Daily at 06:00 IST (Cron)	Ingests updated NSE/BSE security masters, tracks new listings, and maps symbol changes prior to market open.
`evaluate-price-alerts`	Every 15 min (09:15–15:30 IST)	Fetches active alerts, queries current market prices, evaluates trigger conditions, and dispatches Nodemailer alert emails.
`generate-daily-briefing`	Daily at 16:30 IST (Post-Close)	Aggregates day's trading statistics, calculates sector leaders/laggards, calls Gemini AI, and archives post-market brief.

Chapter 7: Verification, Test Suite & Security

Vitest Test Suite Coverage

AUTOMATED CI/CD

- **Formatting Tests:** Validates ₹ INR, Lakh/Crore grouping, and negative numbers.
- **Session State Logic:** Verifies pre-open, normal, closing, and holiday boundaries.
- **Reconciliation:** Tests dual-listed ticker to ISIN resolution and SME flags.
- **Alert Logic:** Evaluates threshold crossing accuracy against simulated feeds.

Security & Compliance

PRODUCTION HARDENED

- **Session Auth:** bcrypt password hashing with secure HTTP-only cookies.
- **Data Isolation:** Strict tenant boundaries on watchlists and personalized alerts.
- **SEBI Compliance:** Explicit non-advisory educational disclaimers and 15-minute delay tags.

Chapter 8: Strategic Future Scope & Roadmap

Dalal Street Pulse is positioned to transition from an analytics and intelligence platform into a comprehensive **Execution, Derivatives & Wealth Management Operating System** for the Indian market.

PHASE 1 1-Click Broker Trading & Execution Gateway

Direct connectivity with SEBI-registered retail stockbrokers via secure OAuth integrations.

- **Broker Adapters:** Seamless integrations with **Zerodha (Kite Connect)**, **Groww**, **Upstox**, **Angel One (SmartAPI)**, and **Dhan HQ**.
- **Native Order Routing:** Place CNC (Long-term), MIS (Intraday), and GTT (Good Till Triggered) orders directly from the stock details terminal.
- **Paper Trading Simulator:** Virtual ₹10 Lakh INR demo portfolio with real-time exchange matching for risk-free strategy validation.
- **Thematic Basket Orders:** Single-click execution of curated stock baskets (e.g. "NIFTY Auto Momentum", "High Dividend PSUs").

PHASE 2 NSE Derivatives (F&O) & Options Analytics Terminal

Professional-grade intelligence for index and equity options traders.

- **Interactive Option Chain:** Live Open Interest (OI), Change in OI, Put-Call Ratio (PCR), and Max Pain visualization for NIFTY, BANKNIFTY, and equity F&O contracts.
- **Real-Time Option Greeks:** Dynamic computation of Delta, Gamma, Theta, and Vega with Implied Volatility (IV) and IV Percentile (IVP) curves.
- **Options Strategy Builder:** Visual payoff diagrams for multi-leg strategies (Bull Call Spreads, Straddles, Iron Condors, Calendar Spreads).
- **Futures Build-up Heatmaps:** Automated real-time classification of Long Build-up, Short Covering, Short Build-up, and Long Unwinding.

PHASE 3 Autonomous Agentic AI & Portfolio Auditor

Proactive, autonomous portfolio intelligence optimizing risk and tax liability.

- **CAS Portfolio Health Auditor:** Drag-and-drop parsing of NSDL/CDSL Consolidated Account Statements (CAS) to diagnose sector concentration risk and fund overlaps.
- **Automated Tax-Loss Harvesting:** Proactive alerts prior to March 31 financial year-end identifying capital losses to offset STCG/LTCG tax liabilities under the Indian Income Tax Act.
- **Concall & Transcripts Intelligence:** Automated AI ingestion of quarterly earnings calls; distills management guidance and margin headwinds into 60-second executive summaries.
- **Multilingual Voice & Chatbot:** Regional Indian language support (Hindi, Gujarati, Tamil, Marathi, Bengali, Telugu).

PHASE 4 IPO Central & Primary Market Intelligence

Comprehensive tracking of Mainboard and SME Initial Public Offerings.

- **Comprehensive IPO Radar:** Track upcoming, open, and closed IPOs with price bands, issue sizes, lot quantities, and anchor investor allocations.
- **Grey Market Premium (GMP) Aggregator:** Real-time estimated listing gains tracking.
- **Live Subscription Telemetry:** Hourly bidding rates across QIB, NII (HNI), and Retail investor quotas.
- **AI DRHP Summarizer:** Instant breakdown of Draft Red Herring Prospectus risk factors, promoter backgrounds, and proceeds utilization.

PHASE 5 Omnichannel Delivery, Native Mobile & Institutional API

Cross-platform ecosystem expansion and B2B data monetization.

- **Native Mobile Applications:** iOS & Android apps (React Native) with biometric authentication and lock-screen live activity tickers.
- **Omnichannel Alerts:** Instant notifications delivered straight to **WhatsApp** and **Telegram** for target prices, corporate actions, and dividend dates.
- **Institutional Developer API:** WebSocket L1/L2 data access, historical tick query endpoints, and white-label analytical widgets for fintech developers.

8.1 Multi-Year Implementation Schedule

Period	Milestone Focus	Deliverables & Technical Capabilities
Q1 – Q2	Broker Execution & Paper Trading	OAuth connectors for Zerodha & Upstox, paper trading virtual engine, custom multi-metric screener.
Q3	Derivatives & Options Hub	Interactive NSE Option Chain, multi-leg strategy payoff visualizer, real-time Greeks & PCR telemetry.
Q4	AI Auditor & IPO Suite	CAS statement parser, March 31 tax-loss harvesting engine, IPO GMP tracker, and DRHP summarizer.
Year 2+	Mobile Apps & Enterprise API	React Native iOS/Android apps, WhatsApp Bot alert integration, and institutional WebSocket API tier.

Chapter 9: Conclusion & Engineering Assessment

Dalal Street Pulse demonstrates that financial intelligence platforms can achieve institutional caliber without sacrificing usability, speed, or domestic market fidelity.

By establishing a robust architectural foundation based on **Next.js 16**, **ISIN canonical data reconciliation**, **Inngest event-driven workflows**, and **grounded valuation AI**, the platform resolves long-standing compromises in Indian retail financial tooling. With the strategic roadmap outlined in this document, Dalal Street Pulse is uniquely positioned to scale into India's premier integrated investment, derivatives, and execution terminal.

Architectural Strengths VERIFIED

- Zero synthetic data: 100% reconciled against verified official exchange masters.
- Resilient offline fallback: Operates flawlessly even without live external MongoDB or API keys.
- End-to-end typed safety: TypeScript strict enforcement across all 27 routes and schemas.

Regulatory Compliance Notice SEBI DISCLOSURE

- Platform is engineered strictly for educational, informational, and analytical intelligence.
- Does not dispense SEBI-registered personalized investment advice or portfolio management.
- All market data feeds comply with statutory 15-minute exchange delay guidelines.